



# David Sinden

APPLIED MATHEMATICIAN — RESEARCH SOFTWARE ENGINEER

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## Professional Experience

### Fraunhofer Institute for Digital Medicine MEVIS

Bremen, Germany

SENIOR RESEARCH SCIENTIST — MODELLING & SIMULATION GROUP — PROF. TOBIAS PREUSSER

2019 - present

- Designed and implemented patient-specific thermal-ablation simulations for microwave, radio-frequency and ultrasound therapies, improving both treatment planning accuracy and computation times
- Developed pharmacokinetic models of liver function and regeneration to support personalized therapy assessment
- Advanced ultrasound beamforming and transcranial imaging methods

### National Physical Laboratory

Teddington, UK

SENIOR RESEARCH SCIENTIST — ULTRASOUND & UNDERWATER ACOUSTICS GROUP — PROF. BAJRAM ZEQRIRI

2014 - 2019

- Modeled piezo- and pyroelectric sensors using multi-physics finite-element simulations to guide device design and characterization
- Developed and maintained computational toolkits for high-resolution ultrasound field mapping and quantitative beam-profile analysis
- Developed measurement-based simulations of nonlinear acoustic propagation in complex media, improving predictive accuracy

### Institute of Cancer Research/The Royal Marsden Cancer NHS Foundation Trust

Sutton, UK

POST-DOCTORAL RESEARCH ASSOCIATE — THERAPEUTIC ULTRASOUND GROUP — PROF. GAIL TER HAAR

2011 - 2014

- Design and implementation of treatment planning software for a large phased-array ultrasound transducer for transcatheter thermal ablation

### University College London

London, UK

POST-DOCTORAL RESEARCH ASSOCIATE — MECHANICAL ENGINEERING — PROF. NADER SAFARI | PROF. ELEANOR STRIDE

2008 - 2011

- Modelling cavitation activity in tissue during high-intensity focused ultrasound therapy

## Education

### PhD — University College London

DYNAMICAL SYSTEMS

2004 - 2008

- Thesis: "Integrability, Localisation and Bifurcation of an Elastic Conducting Rod in a Uniform Magnetic Field"
- Advisor: Prof. Gert van der Heijden

### MSc — University of Bath

MODERN APPLICATIONS OF MATHEMATICS

2003 - 2004

### BSc — Imperial College London

MATHEMATICS WITH APPLIED MATHEMATICS/MATHEMATICAL PHYSICS — 2.1

2000 - 2003

## Skills

|               |           |                      |    |                                                                                                                                                 |
|---------------|-----------|----------------------|----|-------------------------------------------------------------------------------------------------------------------------------------------------|
| SOFTWARE:     | ■ ■ ■ ■ ■ | Python               | 10 | Contributor to open-source projects, open-source code in papers                                                                                 |
|               | ■ ■ ■ ■ ■ | C++                  | 8  | Including STL, Boost, Eigen, VTK and ITK libraries                                                                                              |
|               | ■ ■ ■ ■ ■ | Accelerators         | 4  | OpenCL, numba/cupy, jax/XLA                                                                                                                     |
|               | ■ ■ ■ ■ ■ | Matlab/Octave        | 20 | Was used extensively in research                                                                                                                |
|               | ■ ■ ■ ■ ■ | Fortran              | 14 | Experience from MSc, PhD, post-doc (BLAS, Lapack, auto07)                                                                                       |
| OS:           | ■ ■ ■ ■ ■ | Linux                | 14 | Ubuntu/WSL                                                                                                                                      |
| PRESENTATION: | ■ ■ ■ ■ ■ | -                    | 10 | html/css (tailwind), $\LaTeX$ , Bib $\TeX$                                                                                                      |
| METHODS:      | ■ ■ ■ ■ ■ | Software development | 8  | Version control (git/svn), continuous integration, build systems (CMake, qmake), testing (googletests, pytest), documentation (doxygen, sphinx) |

## Department of Mobility - Constructor University Bremen

*Bremen, Germany*

ADJUNCT LECTURER

*2022, 2024 - present*

- Calculus and Linear Algebra for Graduate Students [MDE-MET-01]: introductory mathematics course for MSc students in data science (2024)
- Numerical Methods [JTMS-MAT-13]: second year mathematics course for physics, engineering and mathematics students (2024-)
- Numerical Analysis [CA-S-MATH-804]: final year course for mathematics students (2022)

## CIMPA Summer School - University of Havana

*Havana, Cuba*

GUEST LECTURER

*June 2023*

- Delivered short lecture course on “Examples and Principles of Mathematical Modelling in Medicine”, to around 50 applied mathematicians from Central and South America and Africa

## Department of Mechanical Engineering - University College London

*London, UK*

SEMINAR TEACHER

*2011*

- Modelling and Analysis in Engineering I [MECH1010]: first year mathematics course for engineering students

## Department of Civil, Environmental & Geomatic Engineering - University College London

*London, UK*

SEMINAR TEACHER

*2008, 2009*

- Mathematics for Engineers II [Math6502]: Second year mathematics course for engineering students

## Nazarbayev University

*Astana, Kazakhstan*

ADMINISTRATION | SEMINAR TEACHER

*2011*

For students on prestigious “Bolashak” scholarship and in establishing partner campus at Nazarbayev University

- Involved in construction of mathematics modules for new courses in mechanical and civil engineering degrees, design of syllabus and preparation of course notes
  - Project management skills, including liaising with host organisation and participating partner institutions from the United States
- Ran tutorials and revision classes for students in mathematics and physics foundation classes
- Marking of tests and exam scripts

## PhD Industrial Supervisor

UNIVERSITY COLLEGE LONDON

*2018–2019*

- Morgan Roberts, Dept. Medical Physics (with Prof. Ben Cox) “Ultrasound Computed Tomography of the Breast”
- Santeri Kaupinmäki, Dept. Medical Physics (with Prof. Simon Arridge) “Inverse Problems for Ultrasound Computed Tomography of the Breast”

HERIOT-WATT UNIVERSITY

*2017–2019*

- Katherine Baker, Dept. Mathematics, (with Prof. Lehel Banjai) “Linear and Nonlinear Wave Equation Models with Power Law Attenuation”

## MSc Secondary Supervisor

CONSTRUCTOR UNIVERSITY, BREMEN

*2019-2021*

- Sandeep Gyawali, Dept. Mathematics, (with Prof. Tobias Preusser). “Extending Composite Finite Element Method for PDE Problems with Geometric Uncertainties”

## Undergraduate Supervision

NATIONAL PHYSICAL LABORATORY

*2015*

- Antoine Lucquiaud, École Normale Supérieure de Cachan, “Boundary Element Methods for Bubble Activity”

UNIVERSITY COLLEGE LONDON

*2012*

- Jade Junqua, ENSEIRB-MATMECA and Bordeaux 1, “Investigating mode conversion and heating around the ribs due to high-intensity ultrasound”

## Outreach

- Scientific consultant on documentary “The healing power of sound” (2014)
- NPL “Scientific Ambassador”: delivered talks at a number of schools and colleges on careers in science as well as demonstrations of experiments relating to objective measurements (2016-2019)

# Affiliations, Awards & Achievements

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## Standardization

- Member of IEC Technical Committee 87: Ultrasonics, Working Group 6 – High Power  
Part of working group of internationally recognised experts writing the technical specification “TS 63900: Measurement-based Simulation in water and complex media”

## Scholarships

- M.Sc. funded by an EPSRC scholarship (2003–4); EPSRC funding was awarded for Ph.D. (2004–7) and post-doctoral work (2014)

## Awards

- Challenge Award: Joint first place in IEEE IUS Challenge on Ultrasound Beamforming with Deep Learning (CUBDL) for “Improving image quality of single plane wave ultrasound via deep learning based channel compounding (2020)”
- Conference Award: Honourable mention for paper “Studying the effect of tissue properties on radiofrequency ablation by visual simulation ensemble analysis” VCBM 2022: Eurographics Workshop on Visual Computing for Biology and Medicine (2022)

## Professional Affiliations

- Member of SIAM, the Society for Industrial and Applied Mathematics (2004–present), associate member of the IMA, Institute of Mathematics and its Applications, (2016–present), and member of the IOP, Institute of Physics, (2008–present)

## Service

- Reviewer for a number of journals (Int. J. Hyperthermia · Ultrasonics · Ultrasound Med. & Biol. · Med. Phys. · Comp. Meth. Prog. Biomed. · J. Open Source Softw. · IEEE Trans. Biomed. Eng. · Phys. Med. Biol. · IEEE Open J. Eng. Med. Biol.), as well as funding agencies (ANR - France, FWF - Austria, Focused Ultrasound Foundation - International).
- Mentor to junior staff at NPL (2015–2019)
- Maintainer in open-source scientific code: **k-wave-python** , available via pypi  
<https://doi.org/10.5281/zenodo.10719461>  150

## Equality, Diversity & Inclusion

- Member of Fraunhofer MEVIS diversity and inclusion task force (2021-)
- Member of ICR's Athena Swan board (2012)

## Grants

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In descending chronological order:

|      |                                                                                                                                                                                                            |          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 2023 | <b>Fraunhofer DISCOVER</b><br>CompTop: Computational Topology in Medical Imaging                                                                                                                           | €150,000 |
| 2023 | <b>European Metrology Programme for Innovation and Research (EMPIR)</b><br>MAIBAI: Developing a Metrological Framework for Assessment of Image-based Artificial Intelligence Systems for Disease Detection | €180,000 |
| 2022 | <b>Fraunhofer-Netzwerk: Simulation</b><br>Physics-Informed Neural Networks                                                                                                                                 | €11,000  |
| 2019 | <b>European Metrology Programme for Innovation and Research (EMPIR)</b><br>RaChy: Radiotherapy Coupled with Hyperthermia – Adapting the Biological Equivalent Dose Concept                                 | £180,000 |
| 2018 | <b>Analysis for Innovators (A4I), with Deltex Medical Devices</b><br>Optimizing Oesophageal Doppler Transducers                                                                                            | £26,500  |
| 2018 | <b>Industrial Challenge Strategy Fund, Wave 1, Metrology for Medical Imaging, with Huntleigh Diagnostics</b><br>Optimizing Fetal Doppler Transducers                                                       | £45,500  |
| 2016 | <b>EPSRC Network+ Therapy Ultrasound Network for Drug Delivery &amp; Ablation Research (ThUNDDAR) grant</b><br>Machine Learning for Cavitation Detection                                                   | £26,500  |
| 2014 | <b>NPL Strategic Research Award</b><br>Mathematical Modelling of Histotripsy                                                                                                                               | £25,000  |
| 2012 | <b>EPSRC/ICR Platform Grant</b><br>Vascular Remodelling                                                                                                                                                    | £25,000  |

- 2024 [David Sinden](#), “Numerical modelling for simulation and planning of focused ultrasound treatments”  
In *Image-guided Focused Ultrasound Therapy: Physics and Clinical Applications*, Eds. Feng Wu, Gail ter Haar, and  
Ian Rivens, Series in Medical Physics and Biomedical Engineering, (CRC Press, Baton Rouge, FL, 2024) ISBN  
9781498711357 —

## Publications & Preprints

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In descending chronological order. Citation data from Google Scholar.

- 2026 Raneigh Sabbagh Gol, Marina Evers, Karl Heimes, Tim Gerrits, Sandeep Gyawali, [David Sinden](#), Tobias Preusser and  
Lars Linsen, “Visual Ensemble Analysis with Deep Learning Prediction for Studying the Effect of Tissue Properties on  
Radiofrequency Ablation”. *Comput. Graph. Forum.* XX, XXX —
- 2025 Christina A. Neizert, Hoang N. C. Do, Miriam Zibell, [David Sinden](#), Christian Rieder, Jakob Albrecht,  
Stefan M. Niehues, Kai S. Lehmann, and Franz G. M. Poch, “Optimizing microwave ablation planning with the  
ablation success ratio”. *Sci. Rep.* 15, 10450 —
- 2022 Pauline Coralie Guillemain, [David Sinden](#), Yacine M’Rad, Michael Schwenke, Jennifer Le Guevelou, Johan Uiterwijk,  
Orane Lorton, Max Scheffler, Pierre-Alexandre Poletti, Jürgen Jenne, Thomas Zilli, and Rares Salomir, “A novel  
concept of transperineal focused ultrasound transducer for prostate cancer local deep hyperthermia treatments”.  
*Cancers* 15, 163 *cites: 5*
- 2022 Christina A. Neizert, Hoang N. C. Do, Miriam Zibell, Christian Rieder, [David Sinden](#), Stefan M. Niehues,  
Janis L. Vahldiek, Kai S. Lehmann, and Franz G. M. Poch, “Three-dimensional assessment of vascular cooling effects  
on hepatic microwave ablation in a standardized ex vivo model”, *Sci. Rep.* 12, 17061 *cites: 9*
- 2022 Karl Heimes, Marina Evers, Tim Gerrits, Sandeep Gyawali, [David Sinden](#), Tobias Preusser, and Lars Linsen, “Studying  
the effect of tissue properties on radiofrequency ablation by visual simulation ensemble analysis”, in *Eurographics  
Workshop on Visual Computing for Biology and Medicine*, Eds. R. G. Raidou, B. Sommer, T. W. Kuhlen, M. Krone,  
T. Schultz, and H-Y. Wu (The Eurographics Association, 2022) ISBN 978-3-03868-177-9, ISSN 2070-5786 *cites: 3*
- 2021 Dongwoon Hyun, Alycen Wiacek, Sobhan Goudarzi, Sven RothlÜbbers, Amir Asif, Klaus Eickel, Yonina C. Eldar,  
Jiaqi Huang, Massimo Mischi, Hassan Rivaz, [David Sinden](#), Ruud J. G. van Sloun, Hannah Strohm, and Muyinatu A.  
Lediju Bell, “Deep learning for ultrasound image formation: CUBDL evaluation framework and open datasets”, *IEEE  
Trans. Ultrason. Ferroelectr. Freq. Control* 68, 3466–3483 *cites: 107*
- 2021 Santeri Kaupinmäki, Ben Cox, Simon Arridge, Christian Baker, [David Sinden](#), and Bajram Zeqiri, “Pyroelectric  
ultrasound sensor model: directional response”, *Meas. Sci. Technol.* 32, 035106 *cites: 5*
- 2020 Sven RothlÜbbers, Hannah Strohm, Klaus Eickel, Jürgen Jenne, Vincent Kuhlen, [David Sinden](#), and  
Matthias Günther, “Improving image quality of single plane wave ultrasound via deep learning based channel  
compounding”, *2020 IEEE International Ultrasonics Symposium (IUS)* (IEEE, 2020) pp. 1–4 *cites: 35*
- 2020 Nadia A. S. Smith, [David Sinden](#), Spencer A. Thomas, Marina Romanchikova, Jessica E. Talbott, and  
Michael Adeogun, “Building confidence in digital health through metrology”, *Br. J. Radiol.* 93, 20190574 *cites: 17*
- 2017 [David Sinden](#), Srinath Rajagopal, N. Christopher Chaggares, Guofeng Pang, and Oleg Ivanytskyy, “Reducing  
uncertainties for spatial averaging at high frequencies”, *2017 IEEE International Ultrasonics Symposium (IUS)* (IEEE,  
2017) pp. 1–4 *cites: 1*
- 2017 Ki Joo Pahk, Pierre Gélât, [David Sinden](#), Dipok Kumar Dhar, and Nader Saffari, “Numerical and experimental study of  
mechanisms involved in boiling histotripsy”, *Ultrasound Med. Biol.* 43, 2848–2861 *cites: 55*
- 2014 [David Sinden](#) and Gail ter Haar, “Dosimetry implications for correct ultrasound dose deposition: uncertainties in  
descriptors, planning and treatment delivery”, *Trans. Cancer Res.* 3, 459–471 *cites: 16*
- 2012 [David Sinden](#), Eleanor Stride, and Nader Saffari, “Approximations for acoustically excited bubble cluster dynamics”,  
*J. Phys.: Conf. Ser.*, Vol. 353 (IOP Publishing, 2012) p. 012008 *cites: 5*

- David Sinden and Gert H. M. van der Heijden, “The buckling of magneto-strictive Cosserat rods”, in Proc. 7<sup>th</sup> European Nonlinear Dynamics Conference (ENOC 2011), edited by D. Bernardini, G. Rega, and F. Romeo (European Mechanics Society, 2011) p. 4, ISBN 978-88-906234-2-4 —
- Gert H. M. van der Heijden and David Sinden, “Localisation of a twisted conducting rod in a uniform magnetic field: the Hamiltonian-Hopf-Hopf bifurcation”, in Proc. 7<sup>th</sup> European Nonlinear Dynamics Conference (ENOC 2011), edited by D. Bernardini, G. Rega, and F. Romeo (European Mechanics Society, 2011) p. 4, ISBN 978-88-906234-2-4 *cites: 2*
- David Sinden, Eleanor Stride, and Nader Saffari, “The effects of nonlinear wave propagation on the stability of inertial cavitation”, *J. Phys.: Conf. Ser.*, Vol. 195 (IOP Publishing, 2009) p. 012008 *cites: 5*
- David Sinden and Gert H. M. van der Heijden, “Spatial chaos of an extensible conducting rod in a uniform magnetic field”, *J. Phys. A: Math. Theor.* 42, 375207 *cites: 10*
- David Sinden and Gert H. M. van der Heijden, “Integrability of a conducting elastic rod in a magnetic field”, *J. Phys. A: Math. Theor.* 41, 045207 *cites: 10*

## Presentations

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In descending chronological order.

- 2025 Nonlinear dynamics of microbubbles in tissue, Dynamical Systems and Geometry Seminary, University of Bremen, 23 January 2025 [\[Abstract\]](#) [\[Presentation\]](#) *Invited*
- 2023 Integrability, localisation and bifurcation of an elastic conducting rod in a magnetic field, 7<sup>th</sup> Workshop on Dynamical Systems & Ergodic Theory in Northern Germany, 9 June 2023 [\[Abstract\]](#) [\[Presentation\]](#) *Invited*
- Artificial intelligence in therapeutic ultrasound, 22<sup>nd</sup> International Symposium on Therapeutic Ultrasound, Lyon, 17–20 April 2023 [\[Abstract\]](#) *Invited*
- 2022 Patient-specific modelling of microwave ablation, Society for Thermal Medicine 2022 Annual Meeting, 1–4 May 2022 [\[Abstract\]](#) [\[Presentation\]](#)
- 2020 Factors for validation of measurement-based simulation, ASA 179, ASA Acoustics Virtually Everywhere, 8 December 2020. [\[Abstract\]](#)[\[Presentation\]](#)
- 2018 Machine learning for cavitation detection, British Medical Ultrasound Symposium, 5 December 2018
- 2016 Acceleration techniques for acoustic holography, British Medical Ultrasound Symposium, 8 December 2016
- Computational challenges in high-intensity focused ultrasound, University of Strathclyde, 25 October 2016 *Invited*
- Absorption of ultrasound by tissue: fractional operators and integral equations Maxwell Institute for Applied Analysis, International Centre for Mathematical Sciences, Edinburgh, 7 October 2016 *Invited*
- Mathematical challenges of high-intensity focused ultrasound, Leslie Comrie Lecture, University of Greenwich, 11 April 2016 *Invited*
- Wave3D: A parallelised three-dimensional nonlinear acoustic wave propagation solver, Anglo-French Physical Acoustics Conference 15, London, 13–15 January 2016
- 2015 Computational challenges in high-intensity focused ultrasound treatment planning, University of Surrey, 15 December 2015 *Invited*
- 2014 Computational challenges in high-intensity focused ultrasound treatment planning, 14<sup>th</sup> International Symposium on Therapeutic Ultrasound, Las Vegas, Nevada, 2–4 April 2014
- 2013 Treatment planning of high-intensity focused ultrasound, Medical Modelling Group, University College London, 30 September 2013 *Invited*
- 2012 The challenges in boundary element modelling for high-intensity focused ultrasound treatment planning, Boundary Integral Equation Methods for High-Frequency Scattering, University of Reading, 25 May 2012 *Invited*

The effects of nonlinear wave propagation on thermal ablation high-intensity focused ultrasound, Department of Electrical Engineering, Stanford University, California, 11 April 2012

*Invited*

- 2011 The buckling of magneto-strictive Cosserat rods, 7<sup>th</sup> European Mechanics Society European Nonlinear Oscillations Conference, Rome, Italy, 24–29 July 2011

Localisation of a twisted conducting rod in a uniform magnetic field: the Hamiltonian-Hopf-Hopf bifurcation, 7<sup>th</sup> European Mechanics Society European Nonlinear Oscillations Conference, Rome, Italy, 24–29 July 2011

Cavitation in tissue under high-intensity focused ultrasound, SIAM Conference on Applications of Dynamical Systems, Snowbird, Utah, 22–26 May 2011

The effect of fluid compressibility on multi-bubble cavitation for high-intensity focused ultrasound, 161<sup>st</sup> Meeting of the Acoustical Society of America, Seattle, Washington, 23–27 May 2011

Modelling cavitation in liver tissue under high-intensity focused ultrasound, British Applied Mathematics Colloquium, University of Birmingham, 11–13 April 2011

Cavitation in models of wave propagation through tissue under high-intensity focused ultrasound, Anglo-French Physical Acoustics Conference 11, Fréjus, France, 19–21 January 2011

- 2010 The influence of liquid viscosity and compressibility on multi-bubble cavitation, UK Therapeutic Ultrasound Interest Group, University College London, 20 December 2010

Multi-bubble interactions, and high-intensity focused ultrasound therapy, 10th International Symposium on Therapeutic Ultrasound, Tokyo, 9–12 June 2010

On the stability of interacting bubbles, UK Therapeutic Ultrasound Interest Group, Institute of Cancer Research, 11 May 2010

Cavitation in high-intensity focused ultrasound treatment, Medical Modelling Group, University College London, 4 May 2010

Integrability, spatially complex localisation and bifurcation of an elastic conducting rod in a uniform magnetic field, London Dynamical Systems Workshop, Imperial College, 29 April 2010

*Invited*

Phase synchronisation and the collective instability oscillating bubble clouds, 159<sup>th</sup> Meeting of the Acoustical Society of America, Baltimore, Maryland, 19–23 April 2010. *J. Acoust. Soc. Am.* 127(3), 1865–1865

On multi-bubble interactions, Anglo-French Physical Acoustics Conference 10, Kendal, 18–22 January 2010

- 2009 On multi-bubble interactions, UK Therapeutic Ultrasound Interest Group, University College London, 11 November 2009

The effects of viscoelasticity on the stability of inertial cavitation, 9<sup>th</sup> International Symposium on Therapeutic Ultrasound, Aix-en-Provence, 23–26 September 2009

- 2008 The effects of nonlinear wave propagation on inertial cavitation, UK Therapeutic Ultrasound Interest Group, University College London, 18 December 2008

The effects of nonlinear wave propagation on inertial cavitation, Anglo-French Physical Acoustics Conference 9, Arcachon, 8–10 December 2008

Integrability, spatially complex localisation and bifurcation of an elastic conducting rod in a uniform magnetic field, University of Surrey, 3 October 2008

*Invited*

Spatially complex localisation of an elastic conducting rod in a uniform magnetic field, Bifurcations in Dynamical Systems with Applications, University of Bielefeld, 19–21 May 2008

- 2007 The integrability of a conducting elastic rod in a magnetic field, British Applied Mathematics Colloquium, Bristol University, 17–19 April 2007